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BrightScript Profiler

You can use the BrightScript Profiler to pinpoint where performance improvements and efficiencies can be made in your Roku app. The tool enables you to collect and analyze the following metrics:

- CPU Usage: Determine where BrightScript code execution is happening
- Wall-Clock time: Determine where the most time (execution and waiting) is being spent when an app is running
- Function call counts
- Memory usage, including memory leak detection

You can use each of these metrics to diagnose problems and get insights where you can improve the app's performance.

> Developers can build a custom profiling tool by following the BrightScript profiler file format specification (/docs/developer-program/dev-tools/brs-profiler-file-format.md). This specification describes the entries in the header, body, and footer of a BrightScript profiler file (.bsprof) file.

Usage

The workflow of the BrightScript Profiler is as below:

1. Add at least the required manifest entries to the app to run the profiler
2. Run and then Exit the app to generate data and metrics
3. Save the profiling data to the device or stream it to the machine (PC) you are using for development
4. Analyze the profiling data as necessary

Manifest entries

Below is the list of manifest keys used by the profiler:

Manifest Entry	Value Type	Legal Values	Default Value	Required	Description
bsprof_data_dest	enum	local , network	local	Yes	If this entry value is `loca` profiling data is collected the device and can be downloaded from the Application Installer afte app terminates. This is th default. If this entry valu `network` , the profiling d is sent over the network rather than being stored the device. See the secti Retrieving Profiling Data details.
bsprof_enable	boolean	0 , 1	0	Yes	Turns on BrightScript Pro when the app is running. is the master flag and mu be set to 1 for any other profiling options to take effect.

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Manifest Entry	Value Type	Legal Values	Default Value	Required	Description
bsprof_enable_lines	boolean	0 , 1	0	Yes	Collects memory and CP data for each line of BrightScript source code makes it easier to pinpoi memory and CPU usage issues. This value is set to 0 by default, which means tha data is collected for each BrightScript function as a whole. Enabling this feature can a significant impact on d performance. Requires BrightScript pro to be enabled (`bsprof_enable=1`), and RSG version to be set to (`rsg_version=1.2`).
bsprof_enable_mem	boolean	0 , 1	0	Yes, if using memory profiling	Turns on memory profilin Requires BrightScript pro to be enabled (`bsprof_enable=1`). If this is enabled, the `bsprof_sample_ratio` is automatically set to 1.0.
bsprof_pause_on_start	boolean	0 , 1	0	No	Immediately after launch the app, profiling is pause until manually resumed w the bsprof-resume comm on the port 8080 debug console. This is useful for profiling isolated parts of an app's or operations, rather tha profiling the entire startu sequence of the app. Requires BrightScript pro to be enabled (`bsprof_enable=1`).

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Manifest Entry	Value Type	Legal Values	Default Value	Required	Description
bsprof_sample_ratio	float	0.001 to 1.0	1.0	Yes	Sets how often profiling samples are taken, while app is running. Only has effect if `bsprof_enable=` memory profiling is enab (`bsprof_enable_mem=1` this value is automaticall to 1.0. The `bs_prof_sample_rat` can be adjusted from 0.0 1.0. A sample ratio of 1.0 the default and provides most accurate data beca every BrightScript stater is measured. A sample ra 1.0, however, may slow dc device performance, but not typically affect the usability of the app. If slo device performance is observed, reduce the rati lessen the profiler's overh
rsg_version	float	1.1, 1.2		Yes, if using line-levl memory profiling	To use line-level profiling, must be set to 1.2 (`rsg_version=1.2`). If it i set to 1.2, profiling will st work correctly; however, level data will not be generated. `rsg_version 1.2` provide significant performance improvements; therefore should set it to 1.2 regarc whether your app is using line-level profiling. See Roku Manifest for mc information (/docs/developer-program/getting-started/architecture/cha manifest.md).

Running the profiler on an app

To initiate the memory profiler, sideload, run, and then exit the app. The profiling data is complete only after the app exits.

Pausing and resuming profiling

Profiling can be paused and resumed at any time. Use the following commands on port 8080 to either pause or resume the memory profiler:

- `bsprof-pause`
- `bsprof-resume`

If the profiler is paused, very little data is written regardless of the data destination. This allows the profiling data (generally, the data relevant to specific parts of an app's UI or other operation) to be collected and analyzed later. These two commands are particularly useful when combined with the `bsprof_pause_on_start` manifest entry.

Manifest entry:

- `bsprof_pause_on_start=1`

For example, if starting video playback is slow or seems to cause memory leaks, the `bsprof_pause_on_start=1` entry can be set in the app's manifest. After the app is launched, but prior to video playback, execute the bsprof-resume command on port 8080 to begin collecting profiling data. After performing the UI operations to be profiled, execute the bsprof-pause command to suspend the storing operation of the profiling data. Then, exit the app to make the profiling data available for analysis. In this scenario, the profiling data will pertain specifically to the operations performed between `bsprof-resume` and `bsprof-pause`.

Port 8080 Commands

These profiling commands exist on port 8080 (Roku OS Versions 9 and later):

Command	Purpose
bsprof-status	Get the status of BrightScript profiling
bsprof-pause	Pause the generation of profiling data
bsprof-resume	Resume the generation of profiling data

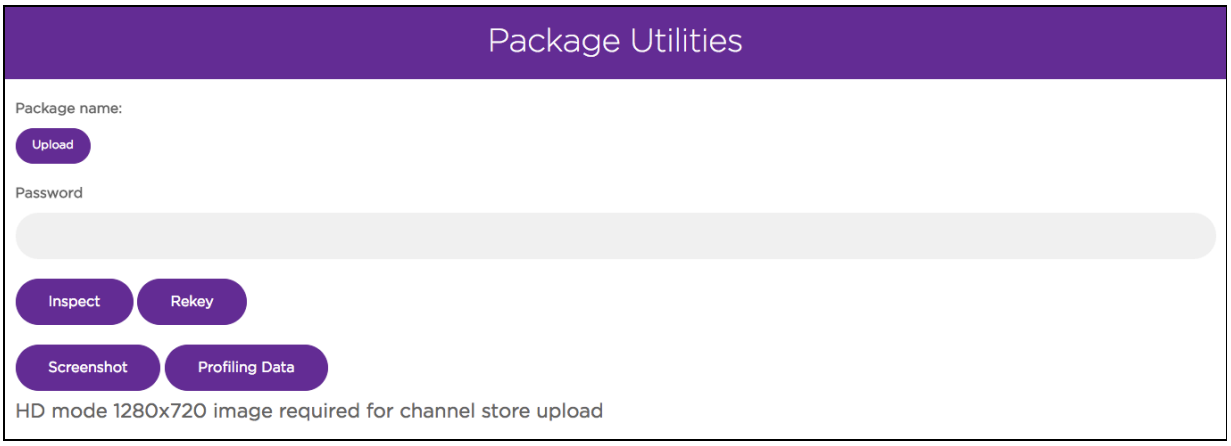
Collecting the data

The app's manifest entry `bsprof_data_dest` determines how the profiling data is retrieved from the device. The data can be stored locally on the device and downloaded after the app finishes running and exits, or it can be streamed over a network connection while the app is running. Streaming consumes significantly less memory on the device while the app is running. In addition, if the app fails, the memory data will have been accurately collected up until the time of the crash (CPU data, however, is typically lost if a failure occurs).

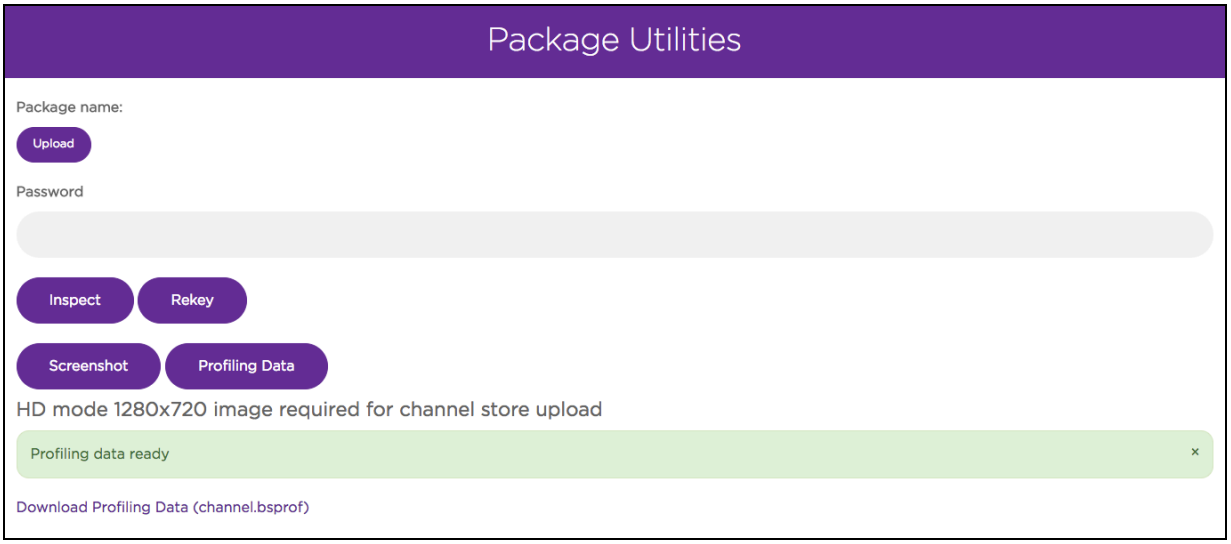
Data Destination: Local

Local data storage is the default storage for profiling, though it can be explicitly selected by adding `bsprof_data_dest=local` to the manifest file. When using this destination, the data becomes available on the device's Application Installer after the app exits:

1. Launch the app and run through the test cases. Once the app finishes running and exits, open Roku device's Developer Settings and click on Utilities.



2. Click Profiling Data to generate a `.bsprof` file and a link to download the data from your Roku device.



The `.bsprof` format is unique to Roku to ensure the format is as efficient and small as possible and easy to generate even on low end Roku devices.

Data Destination: Network

In order to stream an app's profiling data to a network while the app is running, add `bsprof_data_dest=network` to the app's manifest. Streaming data over the network is especially useful when profiling an app's memory usage because all memory operations are included in the profiling data, and the amount of space necessary to store the data can be very large. By streaming the data to a network, the data size is limited primarily by the host computer receiving the data, and not by the available memory on the device itself. Even while streaming the profiling data to the network, there are still additional demands placed on the device's resources while profiling as compared to running an app without profiling. However, the use of resources on the device is significantly reduced.

When this feature is enabled, the start of the app is delayed until a network connection is received by the device, which is the destination for the data. When the app is launched, a message similar to the following appears on the port 8085 developer console:

```
`08-31 23:15:29.542 [scrpt.prof.connect.wait] Waiting for connection to http://192.168.1.1:8090/bsprof/
```

The URL is used with wget, curl, or a web browser. Once a connection is received from one of those programs, the following message appears on the developer console:

```
`08-31 23:15:38.939 [scrpt.prof.connect.ok] profiler connected`
```

When the app exits, the following message appears on the developer console:

```
`08-31 23:16:04.774 [scrpt.prof.save.ok] Profiling data complete, sent via network`
```

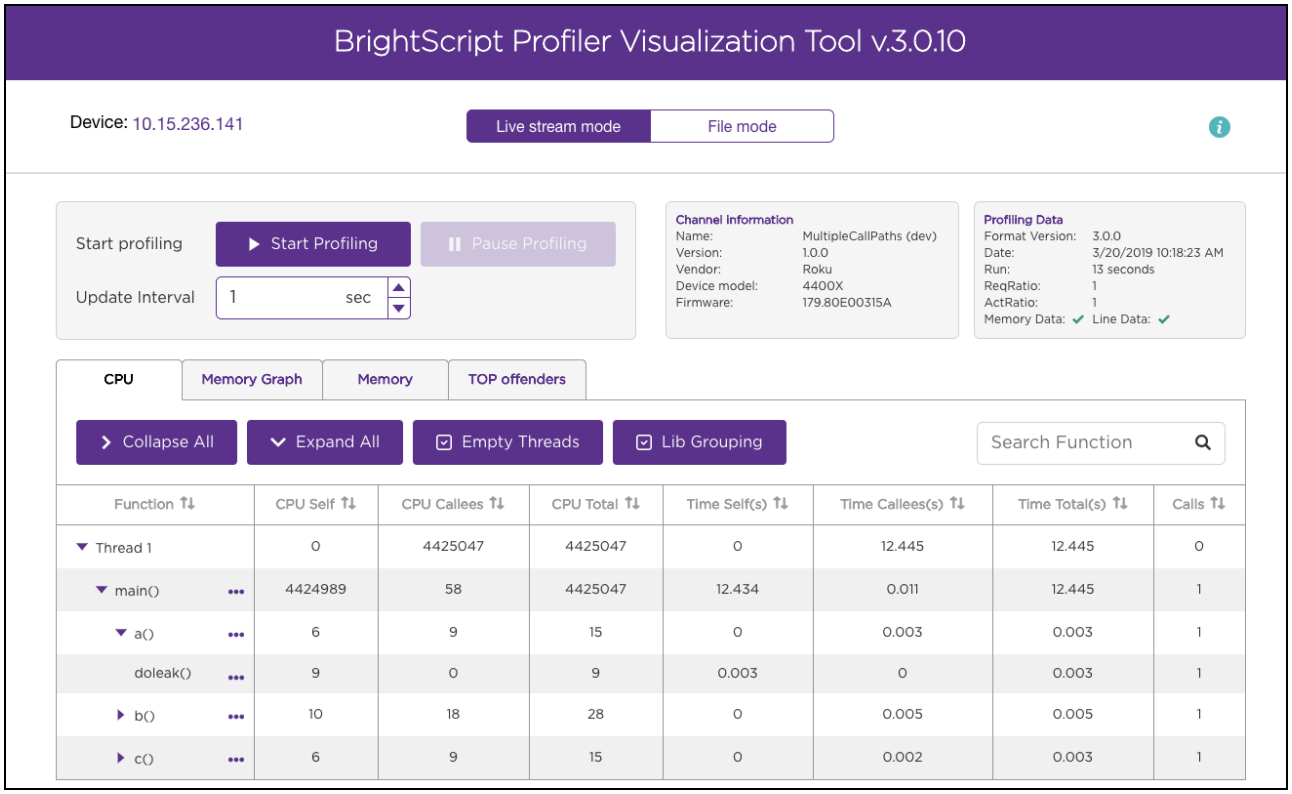
Once that message is seen, the profiling connection is closed by the device and the remote file is populated with profiling data.

Disconnecting from the network during a profiling session may cause device instability.

Viewing profiling data

After downloading the `.bsprof` file, the data can be viewed using the BrightScript Profiler Visualization Tool (http://devtools.web.roku.com/profiler/viewer/). The tool lists the function calls in each thread. For SceneGraph applications, each thread corresponds to either the main BrightScript thread or a single instance of a `<component>`. For example, if you have a Task node that is instantiated multiple times, each instance will appear as a separate thread. The results are the same for any custom `<component>` in the app that is instantiated multiple times. The main BrightScript thread (`Thread main`) is also represented as a single thread even though it has no `<component>`.

For each function listed in the tool, you can expand it to drill down further into its call path and get more detailed data on the function. If you enabled line-level profiling for the app, you can also click on the ellipses to the right of a function to view profiling data for the individual lines of code within a function.



CPU

The CPU tab lists CPU time, wall-clock time, and call count statistics for each function observed and for the function's associated call path (expand a function to view its call path). CPU time refers to the number of operations each function takes to complete; this number should be equal on the low end and high-end Roku devices. Wall-clock time refers to the real world time that a function takes to complete. This value can vary across different Roku devices. For example, a

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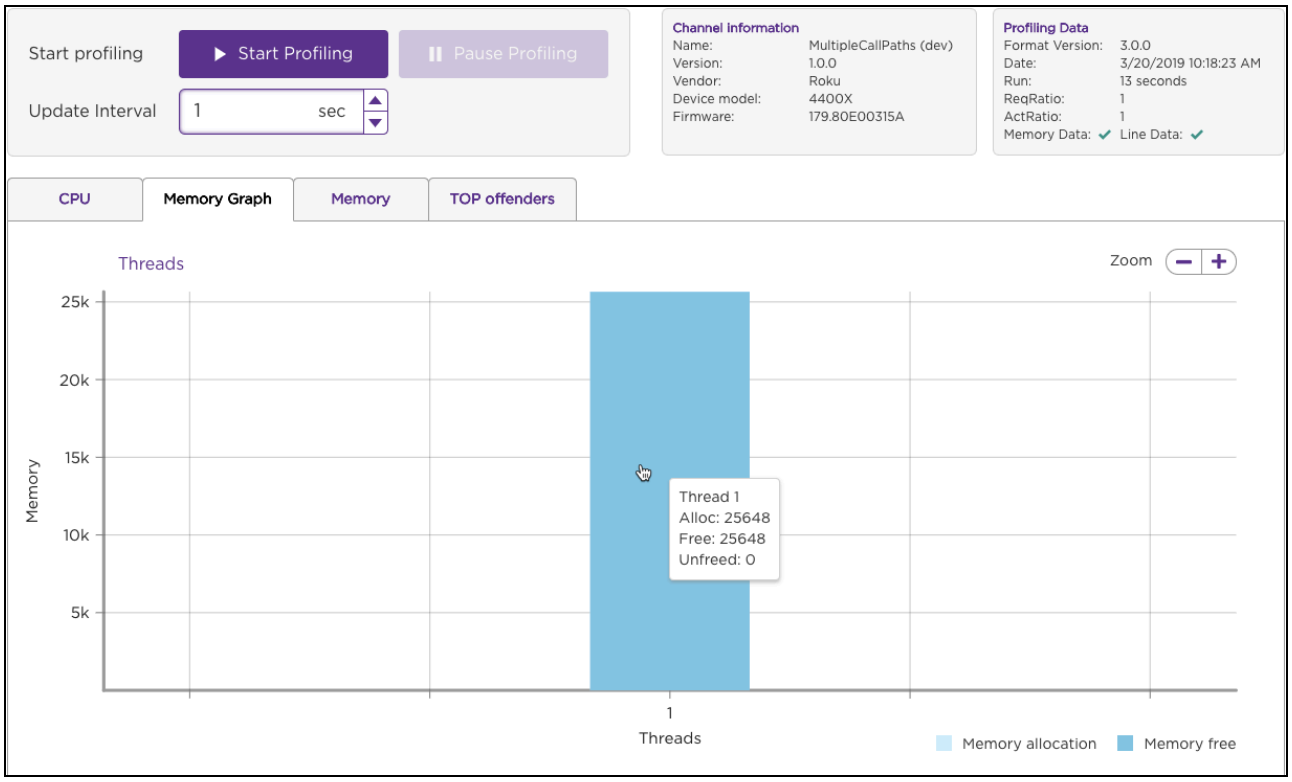
function may take an equal number of operations to complete across different Roku devices but low-end Roku devices can take more real-world time to complete one operation than a high-end Roku device.

The CPU and wall-clock time columns are further divided into separate sections for `self`, `callees`, and `total:`

- self. The CPU/wall-clock time the function consumes itself.
- callees. The amount of CPU/wall-clock time consumed by any functions called by the original function.
- total. The amount of CPU/wall-clock time consumed by the original function (self) and any callee functions.

Memory Graph

The Memory Graph tab visualizes the memory consumption for each thread in the app. Move the mouse pointer over a column to view the allocated, free, and unfreed memory for the thread.



Memory

The Memory tab lists allocated, freed, and unfreed memory statistics for each function observed (self) and for the function's associated call path (total).See Profiling Values for more information on the memory statistics displayed in this tab.

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CPU | Memory Graph | Memory | TOP offenders

Back To Tree Thread 1 › main › b › doleak

Line / Function	Alloc Self	Free Self	Free Other	Unfreed Self	Alloc Total	Free Total	Unfreed Total
28	80	80	0	0	80	80	0
29	6256	6256	0	0	6256	6256	0

Line-Level profiling

You can collect profile data for each line of BrightScript source code to better pinpoint high CPU and memory usage. To do this, enable profiling (`bsprof_enable=1`), line-level profiling (`bsprof_enable_lines=1`), and RSG version 1.2 (`rsg_version=1.2`) in the manifest. In the CPU or Memory tab, click on the ellipses to the right of the function to drill down into its individual lines of code.

CPU | Memory Graph | Memory | TOP offenders

Collapse All Expand All Empty Threads Lib Grouping Search Function

Function	Alloc Self	Free Self	Free Other	Unfreed Self	Alloc Total	Free Total	Unfreed Total
Thread 1	0	0	0	0	25648	25648	0
main()	136	136	0	0	25648	25648	0
a()	40	40	3768	0	6368	6368	0
b()	80	80	7536	0	12752	12752	0
doleak()	6336	6336	0	0	6336	6336	0
doleak()	6336	6336	0	0	6336	6336	0
c()	40	40	3768	0	6392	6392	0

The BrightScript Profiler lists the CPU or memory usage at the time the individual lines of code in the function were executed.

CPU | Memory Graph | Memory | TOP offenders

Back To Tree Thread 1 › main › b › doleak

Line / Function	Alloc Self	Free Self	Free Other	Unfreed Self	Alloc Total	Free Total	Unfreed Total
28	80	80	0	0	80	80	0
29	6256	6256	0	0	6256	6256	0

See Profiling Values for more information on the CPU and memory statistics displayed in the tool.

Profiling Values

The following table describes the profile statistics displayed in the BrightScript profiler tool. The descriptions are applicable for the functions as a whole or for individuals lines of code in the functions (if using line-level profiling).

Values from memory profiling

Calls	Number of times a function was called
Cpu.self	CPU* used in a function, itself
Cpu.callees	CPU* used in functions called by a function
Cpu.total	Cpu.self + cpu.callees
Mem.self	Memory allocated within a function itself, but not freed (leaks)
Mem.callees	Memory allocated by functions called by a function, but not freed (leaks)
Mem.total	Mem.self + mem.callees
Tm.self	Real (wall-clock) time spent on a function, itself
Tm.callees	Real (wall-clock) time spent on functions called by a function
Tm.total	Tm.self + tm.callees
Avg.cpu_self, Avg.cpu_callees, Avg.cpu_total, Avg.mem_self, Avg.mem_total, Avg.tm_self, Avg.tm_callees, Avg.tm_total,	Average of the metric, over the number of calls (e.g., if cpu.self=100 and calls=2, avg_cpu_self will be 50)

A “memory leak” is simply any memory that is allocated, but not freed while the profiler was running. If memory is freed while profiling is paused, the free memory is not tracked and the memory may show up as “leaked.”

Time is measured as if a stopwatch were used to time the action. For example, if a function makes a network call, there may be very little CPU time used, but a significant amount of time waiting for the network response.

If any of these metrics appear in a call path, they are specific to that call path. For example, in this call path:

```
<root>: cpu.self=0,cpu.callees=14700,tm.self=0.000,tm.callees=1.989,mem.self=0,mem.callees=324452,ca
+- func1(): pkg:/components/file1.brs:83,cpu.self=200,cpu.callees=14500,tm.self=0.728,tm.callees=1.261,
| +- func2(): pkg:/components/file2.brs:22,cpu.self=14500,cpu.callees=0,tm.self=1.261,tm.callees=0.000,
```

The metrics for func2() are specific to when it is called from func1().

However, in the table below:

```
------------- BEGIN: TOP CONSUMERS: CPU.SELF -----------------

  1: func1(): pkg:/components/file1.brs:83,cpu.self=300,cpu.total=450,tm.self=0.001,tm.total=0.001,mem.

  2: func2(): pkg:/components/file2.brs:22,cpu.self=55430,cpu.total=80500,tm.self=0.126,tm.total=0.126,

-------------- END: TOP CONSUMERS: CPU.SELF -----------------
```

The metrics displayed are the totals for all calls to each function, on any call path.

Top Offender

The Top Offenders tab displays the functions or function call paths that used the greatest amount of CPU, memory, or time. This tab is typically the best place to start diagnosing potential performance issues. The top 10 functions with the most unfreed memory are listed by default. You can change the report to a top 20, 100, 500 list or enter a custom depth, and you can select which metric to view. For example, the top 5 methods with the highest total CPU time are displayed in the following image:

Function	Unfreed total	Show in the tree
1 main()	0	show call paths
2 a()	0	show call paths
3 doleak()	0	show call paths
4 b()	0	show call paths
5 c()	0	show call paths

Click Show call paths to open the CPU or Memory tab with more detailed profiling data for the selected function and metric.

Using this data

Here are a few key points on how to use this data to improve app performance:

Data Type	Definition and Best Use
High wall-clock time but low CPU time	This pattern shows a function is consistently waiting, whether it be for input or a response from an external source. These functions are best suited for Task nodes so that it doesn't block the main thread.
Complex functions	Try to simplify the as much as possible. If a function handles multiple tasks, consider breaking it out into several functions to further isolate how much CPU or wall-clock time is consumed by each task.
Functions that consume a large amount of CPU or wall-clock time	Move functions to Task nodes, if they are consistently waiting. A function can be determined to be waiting if it's wall-clock time is high, but its CPU cost is low

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